

■ AgriTech — Mandi-to-Market Supply Chain Optimizer

Presentation & Cross-Question Prep Notes
TransOrg AgentIQ Datathon — Track 3

1. One-Line Pitch

"We built a full-stack agricultural supply chain intelligence platform that rescues dirty real-world data, validates it with a rigorous analytics layer, and makes it explorable through an interactive dashboard and a natural-language AI agent."

2. Presentation Story Arc

Problem → **Data Challenges** → **What We Built** → **How It Works** → **Results/Metrics** → **Demo** → **Bonus AI Agent**

- **Do not** start with 'I used Streamlit' — start with the PROBLEM.

3. The Problem (Open Strong)

- Agricultural supply chains involve hundreds of Mandis — data is fragmented, multilingual, and inconsistent in units.
- No single executive view exists to monitor arrivals, price vs MSP, transport efficiency, and weather correlations together.
- **3,924 price crashes** where farmers were paid below the government MSP guarantee — with no automated detection.
- Transport delays affect 18.68% of trips, with no route-level visibility.

4. Your 4-Phase Approach

Phase	What You Did
Data Engineering	Cleaned 5 raw datasets, standardized units, resolved multilingual names, flagged invalid records
Analytics Layer	Built SQLite analytical views to compute KPIs from clean data
Metric Validation	Ran validate_metrics.py to verify every metric against ground truth
Dashboard + AI Agent	Built Streamlit dashboard + natural language query engine on top

5. Dataset Summary — Know This Cold

Dataset	Raw	Clean	Key Issue Fixed
Mandi Master	60	57	Removed 3 exact duplicates
Crop Arrivals	25,750	25,000	Hindi/English names mapped, units → Quintals
Prices & MSP	12,000	12,000	Currency symbols stripped, 638 districts recovered via join
Transport Logistics	10,400	10,000	400 duplicates removed, 539 invalid negative transit times flagged

Weather Sensors	15,000	15,000	°F→°C, inches→mm, UTC→IST
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6. Validated Metrics — Memorize These

Metric	Value
Total Arrivals	5,659,184.82 QTL
Active Mandis	57
Crops Tracked	6
Price Records	12,000
Price Crashes (below MSP)	3,924
Missing Modal Prices	610
Missing MSP Values	2,377
Valid Trips	8,455
Invalid Negative Transit Records	539
Delayed Trips	1,584
Transport Delay Rate	18.68%
Weather-Matched Arrival Days	6,345
Weather-Unmatched Arrival Days	6,001

7. Key Design Decisions — Be Ready to Justify

Decision	Your Answer
Why SQLite?	Zero-config, portable, ships with codebase. No infrastructure overhead.
Why Streamlit?	Fastest Python-native dashboarding. No frontend code required.
Why Newsprint theme?	Executive product deserves editorial authority, not toy colors.
Why regex agent, not LLM?	Deterministic, fast, no API cost, no hallucinations, covers the domain fully.
Why not use abs() on negatives?	False data is worse than missing data. -3hr → +3hr is factually incorrect.
Why not impute MSP nulls?	MSP is a government-policy value. Statistical imputation would be misleading.

8. Delay Methodology

"A trip is classified as delayed when its valid transit time exceeds 1.5× the average valid transit time for that specific Mandi → Warehouse route. Invalid/negative records are excluded from the baseline entirely."

- It is **route-specific**, not a global threshold.
- Prevents bad records from skewing the baseline average.

9. Weather Methodology — Avoid the Causality Trap

If asked 'Does rainfall cause drop in arrivals?' — say:

"Our methodology establishes an association/correlation between rainfall and arrival volumes — not causation. We aggregated weather to daily grain before joining to arrival data to prevent row inflation."

10. Dashboard Sections Guide

Section	Why It Matters
KPI Strip	Executives see 4 headline numbers instantly at a glance
Supply Intelligence	Identifies volume patterns and top-performing Mandis
Price Discovery	Catches farmer distress price crash events
Logistics	Pinpoints supply chain bottlenecks by route
Weather × Supply	Supports procurement and logistics planning

11. AI Query Agent — Explain Technically

"The agent parses the user's question using regex patterns to extract intent (crop, mandi, metric, time range). It dynamically generates SQL against the SQLite analytics views, selects the best chart type, renders a Plotly chart, and provides a text summary."

Sample demo queries:

- 'Show me the daily arrival trend of Wheat'
- 'What are the top 5 mandis by arrival volume?'
- 'Show price crashes for Cotton'
- 'What is the correlation between rainfall and arrivals?'

12. Cross-Questions with Answers

On Data Engineering

Q: How did you handle Hindi crop names?

Explicit mapping dictionary. Names like 'Gehun', 'Kanak', '?????' all map to 'Wheat' based on manual inspection and domain knowledge.

Q: How did you recover 638 missing district values?

Left join between the price dataset and mandi master on mandi_id. Where district was null in price records but available in the master, we backfilled it.

Q: Why retain NULL values instead of filling with 'Unknown'?

Imputation introduces bias. Nulls are transparent and honest — they tell the viewer the data point is genuinely missing.

Q: Why not impute the 2,377 missing MSP values?

MSP is a government-declared policy value. Imputing it with averages might classify a real price crash as normal. We preserved nulls.

On Analytics

Q: How did you define a 'price crash'?

Any record where modal_price < msp — i.e., where the actual wholesale price paid fell below the government's Minimum Support Price guarantee.

Q: What are 'analytical views' in your database?

SQL views built on top of cleaned base tables. Each view pre-joins and pre-aggregates data for specific use cases so dashboard queries stay fast and simple.

Q: How did you validate your metrics?

validate_metrics.py independently recalculates every core metric from clean CSVs and compares against database query results. It serves as a regression test.

On the Dashboard

Q: How does the cross-filtering work?

All filters are applied server-side in Python before data is passed to charts. Streamlit's reactive re-run model means every interaction triggers full recomputation.

Q: Why not use Power BI or Tableau?

Streamlit keeps everything in Python — same language as the pipeline. No separate BI licences, no data extraction step, and the entire solution is version-controlled.

On the AI Agent

Q: Why regex and not an LLM like GPT-4?

Reliability and cost. Deterministic SQL generation is preferable over probabilistic LLM output for a production analytics tool. No API dependencies.

Q: What happens if the user asks something unrecognized?

The agent returns a graceful error message indicating it couldn't parse the query, and suggests example queries the user can try.

Q: Can the agent access raw data?

It only queries the analytics database — not raw CSVs. Intentional, for both performance and data governance reasons.

On Architecture & Scalability

Q: How scalable is this architecture?

For this scope SQLite is adequate. For production, swapping to PostgreSQL or DuckDB is straightforward — the SQL views are standard ANSI SQL. Streamlit can be containerized.

Q: What would you improve with more time?

True LLM integration with a local open-source model, price crash early-warning predictive model, automated alerts when delay rates exceed thresholds, real-time data ingestion.

13. Demo Strategy

1. Start with the KPI strip — anchor the big numbers in the judges' minds.
2. Supply Intelligence — show the arrival trend chart.
3. Price Discovery — zoom in on a price crash cluster.
4. Logistics — highlight a high-delay route.
5. AI Agent — ask 'What are the top 5 mandis by arrival volume?' and show the chart output.

6. Briefly mention the data cleaning pipeline as the foundation that makes everything possible.

Golden Rule: Demo what works. Do not live-demo something fragile.

14. One-Liner Cheat Sheet

Question	Answer
Why SQLite?	Zero-config, portable, adequate for this dataset size.
Why Streamlit?	Fastest Python-native dashboarding without frontend code.
Why Newsprint theme?	Executive product deserves editorial authority, not toy colors.
Why regex agent?	Deterministic, fast, no API cost, covers the domain completely.
Why not fix negatives with abs()?	False data is worse than missing data.
Why MSP nulls not imputed?	Government-policy values cannot be statistically imputed.
Why not LLM?	Reliability and determinism over probabilistic output.

Good luck! Lead with the 3,924 price crashes — it's the most compelling number in the project.